

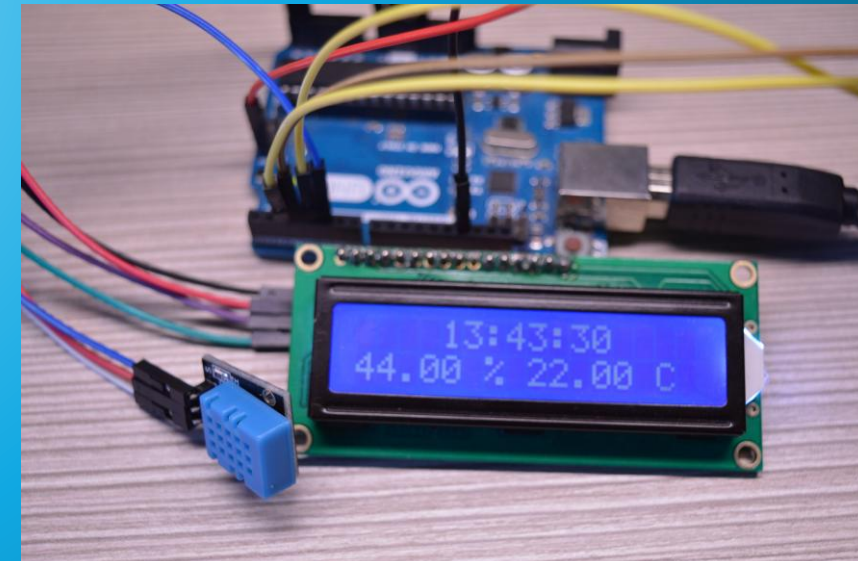
Arduino Weather Station

ECS3001

Project Guide: Prof. Ramesh Prasad Panda

Team Members

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Project Title:

Arduino Weather Station for Real-Time Environmental Monitoring

Overview:

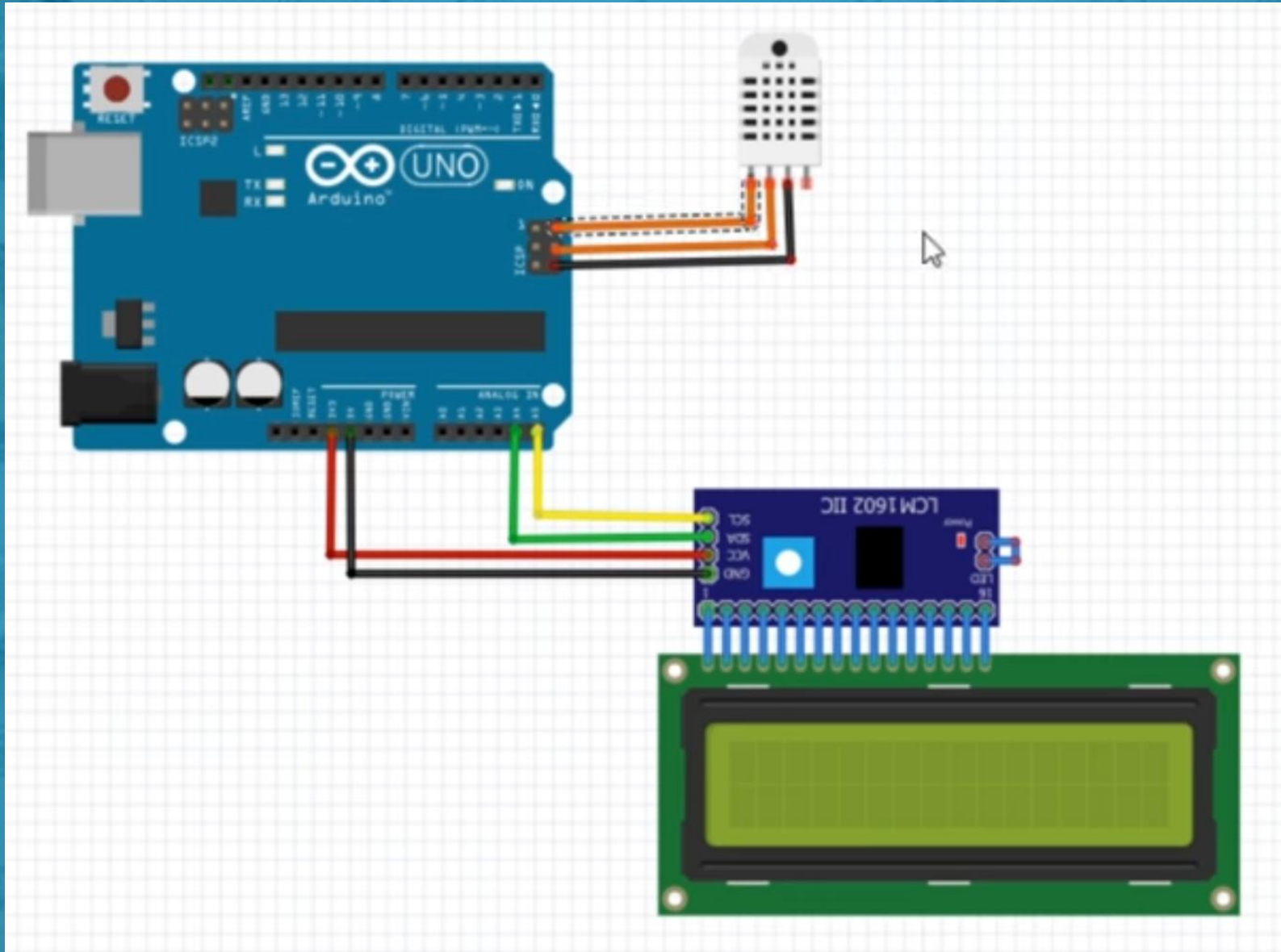
This project focuses on the development of an Arduino-based Weather Station designed to monitor and display real-time environmental parameters. The system collects data such as temperature, humidity, atmospheric pressure, and light intensity using dedicated sensors interfaced with an Arduino microcontroller. The measured data is processed and displayed on an LCD for easy observation.

The weather station provides a low-cost and efficient solution for local weather monitoring and basic climate analysis. By integrating multiple sensors with the Arduino platform, the system demonstrates practical applications of embedded systems and sensor-based data acquisition. The modular design allows easy expansion, such as adding wireless communication or data logging for long-term analysis.

Key Features:

1. Real-Time Weather Monitoring
2. Multi-Sensor Data Acquisition
3. Low-Cost and Modular Design

Circuit-Wiring Diagram:



Equipment and Device Required:

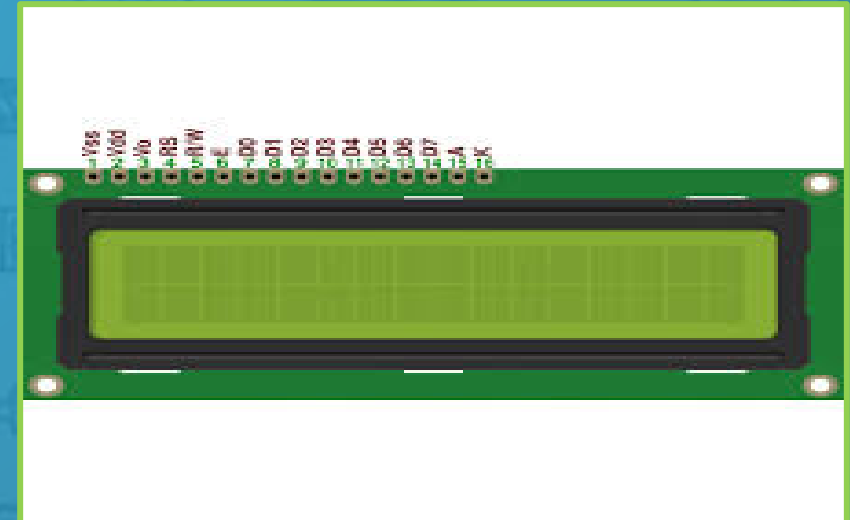
1. Arduino Uno:

The Arduino Uno acts as the main control unit of the weather station. It reads data from various environmental sensors, processes the information, and controls the output devices such as the display. Its ease of programming, wide community support, and multiple I/O pins make it ideal for real-time weather monitoring applications.



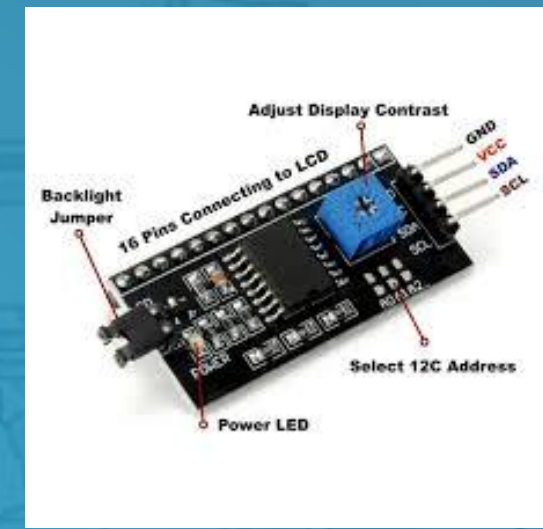
2. 16×2 LCD Display:

The 16×2 LCD display is used to present real-time weather data such as temperature, humidity, and other parameters in a clear and readable format. When used with an I2C module, it reduces wiring complexity and allows efficient communication with the Arduino Uno.



3. I2C module:

The I2C module is used to interface the 16×2 LCD display with the Arduino Uno using only two communication lines (SDA and SCL). This reduces the number of connecting wires, simplifies the circuit design, and improves reliability.



4. DHT11 Sensor:

The DHT11 sensor is used to measure temperature and humidity. It provides digital output data, making it easy to interface with the Arduino Uno. The sensor is low-cost, reliable, and suitable for basic weather monitoring applications.



5. Battery:

A battery is used to supply electrical power to the Arduino Uno, DHT11 sensor, and LCD display. It ensures continuous operation of the weather station by providing a stable power source, allowing the system to measure and display temperature and humidity without interruption, even when external power is not available.



Estimated Cost of Project:

Item	Description / Notes	Approx. Cost (INR)
Arduino Uno	Main controller board (original or compatible)	₹400
16×2 LCD Display	Used to display temperature and humidity	₹200
I2C Module	Reduces wiring for LCD (SDA & SCL only)	₹100
DHT11 Sensor	Measures temperature and humidity	₹150
Jumper Wires	For circuit connections	₹100
Breadboard	For prototyping connections	₹100
Power Supply / USB Cable	Powering the Arduino	₹100
Item	Description / Notes	Approx. Cost (INR)

Total: 1250

Plan of Action:

Design & Planning – Finalize the project objective, select components (Arduino Uno, DHT11 sensor, LCD, I2C module), and prepare the block diagram.

Component Procurement – Collect all required components such as the Arduino board, sensors, display unit, jumper wires, and power supply.

Circuit Assembly – Connect the DHT11 sensor and 16×2 LCD (via I2C) to the Arduino Uno using proper wiring and pin configuration.

Software Development – Write and upload the Arduino program to read temperature and humidity values from the DHT11 sensor.

Display Integration – Interface the LCD with the Arduino and display real-time temperature and humidity readings.

Debugging & Optimization – Identify and fix errors in wiring or code to ensure stable and reliable performance.

Final Demonstration & Documentation – Demonstrate the working model and prepare the project report and presentation.

Timeline Of Progress:

Week	Task	Outcome
Week 1	Finalize project concept, components, and block diagram	Clear project plan
Week 2	Procure Arduino Uno, DHT11 sensor, LCD, and I2C module	All components available
Week 3	Circuit design and wiring connections	Proper hardware setup
Week 4	Write and upload Arduino code	Sensor data successfully read
Week 5	Interface LCD with I2C and display readings	Real-time display working
Week 6	Testing and calibration of temperature & humidity values	Accurate readings obtained
Week 7	Debugging and system optimization	Stable system performance
Week 8	Final demonstration and documentation	Project completed

Thank You